

PLANT REBUILD CENTER

GEAR PUMP

SAR(4)180+180+(1)6

RECEIVING

CHECK SHEET

MACHINE MODEL

HD 785-7

WO NO:

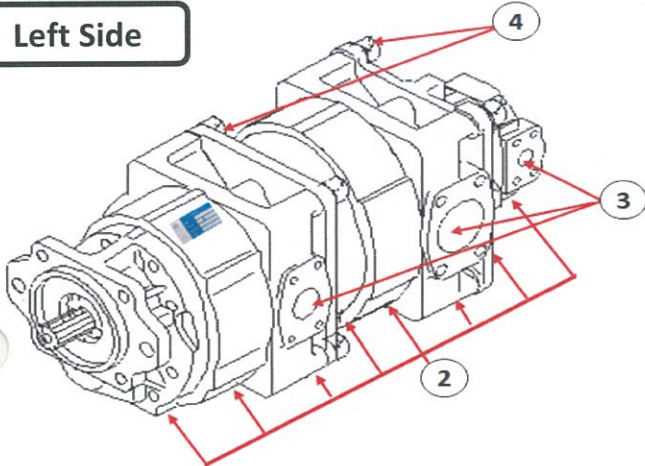
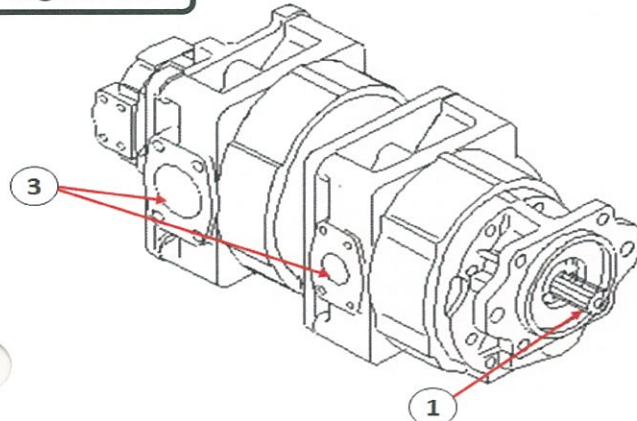
POWER TRAIN SECTION

RECEIVING CHECK SHEET GEAR PUMP SAR (4) 180 + 180 + (1)6	WO. NO		DOCUMENT NO	
	COMP.S/N		PUBLIC DATE	
	INSPECTION DATE		REVISION NO	
	JOB SITE		PAGE	

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM	CHECK	REMARK						
Left Side 	1	Spline of shaft								
	2	Main housing								
	3	contact surface O-ring and treath (Discharge)(Suction)								
	4	Mounting bolt								
		Name plate								
		Painting condition								
		Packing / wraping								
		Stand								
		Bolt stand								
Right Side 										
NOTE : PLEASE MARK "✓" FOLLOWING TO THE ACTUAL CONDITION										
<table border="1"> <tr> <td>OIL LEVEL</td> <td>EMPTY OR DRAINED</td> <td></td> </tr> <tr> <td></td> <td>PROPER LEVEL</td> <td></td> </tr> </table>		OIL LEVEL	EMPTY OR DRAINED			PROPER LEVEL				
OIL LEVEL	EMPTY OR DRAINED									
	PROPER LEVEL									

Note :

INSPECTION BY

CHECK BY

CHECK BY

APPROVED BY

QA INSPECTOR

QA COORDINATOR

GL/SPV PROD

PPC

REV. 00



PLANT REBUILD CENTER

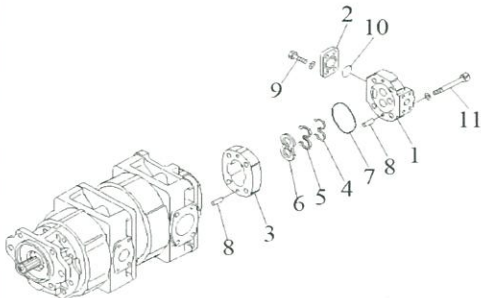
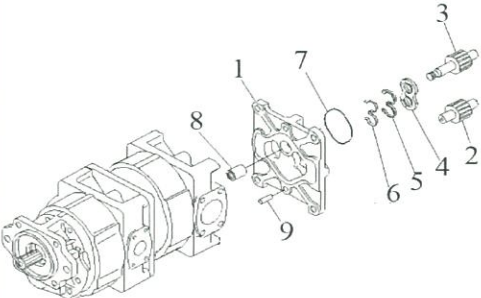
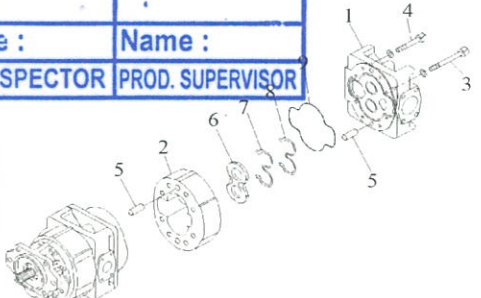
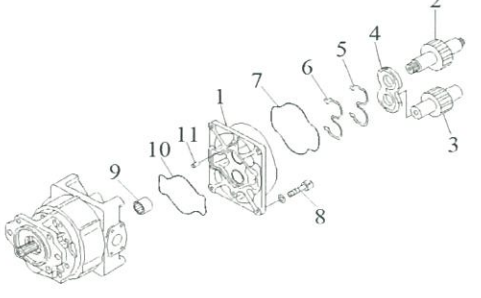
GEAR PUMP
SAR(4)180+180+(1)6
DISSASSEMBLY
CHECK SHEET

MACHINE MODEL

HD785-7

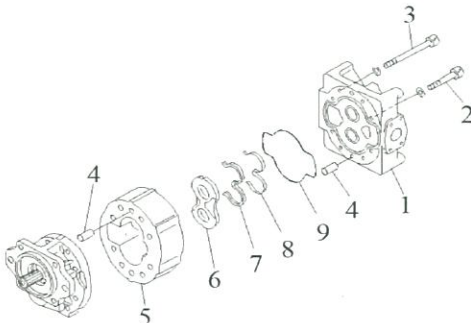
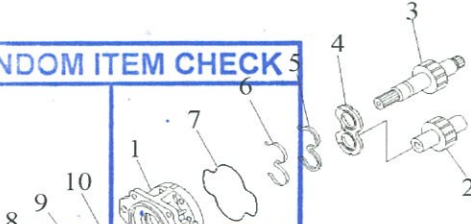
WO NO:

POWER TRAIN SECTION

DISASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO		
GEAR PUMP		COMP.MODEL	PUBLIC DATE		
SAR(4)180+180+(1)6		COMP.S/N	REVISION NO		
		UNIT MODEL	PAGE		
PROCESS	START FINISH	DISASSEMBLY BY			
NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	REUSE SALVAGE REPAIR	SKETCH DRAWING OR PHOTOES	REMARK
	SAR (1)6 DISASSEMBLY	⚠ IMPORTANT ⚠ Check contact surface cover and bracket,treath,and abnormal worn of pin dowel.Check abnormal worn of gear and side plate contact surface gear don't mix • Remove bolt (11)from bracket (1). •Tracking out housing (1) by hammer slowly. • Check dowel pin (8) & holes • Remove O-ring (7) from cover (1) • Remove side plate assy (6) ring back up (5) and seal oil (4) • Remove gear (2) and gear (3) • Remove cradel (3) by bloe hammer • Remoeve pin dowel (8) from cradel (3) • Remove side plate assy (4) ring back up (5) and seal oil (6) • Remove O-ring (7) from bracket (1) • Remove bolt from bracket (1) • Remoeve bracket (1) from body • Remove pin dowel (9) • Remove caupling (8) from body		 	
2	SAR 180 DISASSEMBLY	⚠ IMPORTANT ⚠ Check contact surface cover and bracket,treath,and abnormal worn of pin dowel.Check abnormal worn of gear and side plate contact surface gear don't mix • Remove bolt (3)(4)from bracket (1). •Tracking out housing (1) by hammer slowly. • Check dowel pin (5) & holes • Remove O-ring (9) from cover (1) • Remove side plate assy (6) ring back up (7) and seal oil (8) • Remove gear (2) and gear (3) • Remove cradel (2) by bloe hammer • Remoeve pin dowel (5) from cradel (2) • Remove side plate assy (4) ring back up (5) and seal oil (6) • Remove O-ring (7) from bracket (1) • Remove bolt (8) from bracket (1) • Remoeve bracket (1) from body • Remove pin dowel (11) and O-ring (10) • Remove caupling (9) from body		RANDOM ITEM CHECK Name : _____ Name : _____ QA INSPECTOR PROD. SUPERVISOR  	

DISASSEMBLY CHECK SHEET GEAR PUMP SAR(4)180+180+(1)6	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

PROCESS	START		DISASSEMBLY BY	
	FINISH			

NO	PART TO REMOVE OR INSPECT	CONDITIONS TOBE INSPECTION	REUSE	SALVAGE	REPAIR	SKECTH DRAWING OR PHOTOES	REMARK				
	SAR (4) 180 DISASSEMBLY	⚠ IMPORTANT ⚠ Check contact surface cover and bracket,treath,and abnormal worn of pin dowel.Check abnormal worn of gear and side plate contact surface gear don't mix • Remove bolt (3)(2)from bracket (1). •Tracking out housing (1) by hammer slowly. • Check dowel pin (4) & holes • Remove O-ring (9) from cover (1) • Remove side plate assy (6) ring back up (7) and seal oil (8) • Remove gear (2) and gear (3) • Remove cradel (5) by bloe hammer • Remoeve pin dowel (4) from cradel (5) • Remove side plate assy (4) ring back up (5) and seal oil (6) • Remove O-ring (7) from bracket (1) • Remove plate (10) from bracket (1) • Remoeve oil seal (9) from body • Remove pin dowel (11) and O-ring (10) • Remove snap ring (8) from body				  RANDOM ITEM CHECK <table><tr><td style="width: 50%;">Name :</td><td style="width: 50%;">Name :</td></tr><tr><td>QA INSPECTOR</td><td>PROD. SUPERVISOR</td></tr></table>	Name :	Name :	QA INSPECTOR	PROD. SUPERVISOR	
Name :	Name :										
QA INSPECTOR	PROD. SUPERVISOR										

NOTE

INSPECTION BY

APPROVE BY

MECHANIC

GROUP LEADER (GL)

REV. 00



PLANT REBUILD CENTER

GEAR PUMP

SAR(4)180+180+(1)6

MEASUREMENT

CHECK SHEET

MACHINE MODEL

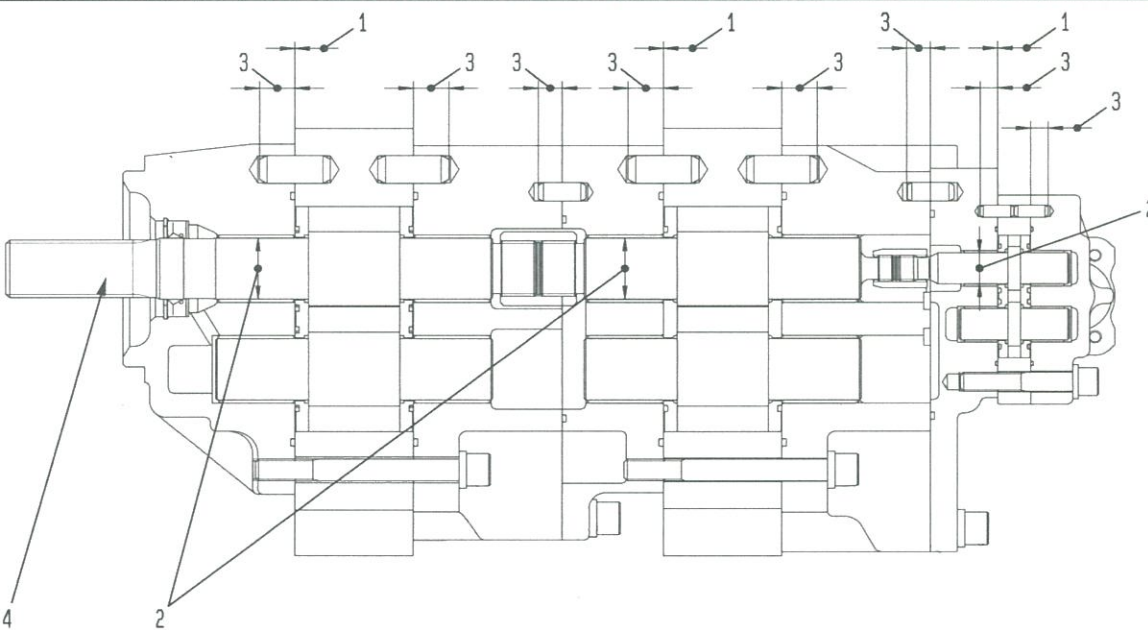
HD785-7

WO NO:

POWER TRAIN SECTION

MEASUREMENT CHECK SHEET GEAR PUMP SAR(4)180+180+(1)6	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

PROCESS	START FINISH		MEASUREMENT BY	
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NO	PART TO MEASURING OR INSPECT	STANDARD VALUE TABLE																																																																									
1	SAR(4)180+180+(1)6 MEASUREMENT	 Unit: mm <table border="1"> <thead> <tr> <th>No.</th><th>Check item</th><th colspan="3">Criteria</th><th>Remedy</th></tr> <tr> <th></th><th></th><th>Model</th><th>Standard clearance</th><th>Clearance limit</th><th></th></tr> </thead> <tbody> <tr> <td>1</td><td>Side clearance</td><td>SAR(4)-180</td><td rowspan="2">0.11 – 0.16</td><td rowspan="2">0.20</td><td rowspan="6">Replace</td></tr> <tr> <td></td><td></td><td>SAR(4)-180</td></tr> <tr> <td></td><td></td><td>SAR(1)-6</td><td>0.10 – 0.15</td><td>0.19</td></tr> <tr> <td>2</td><td>Clearance between inside diameter of plain bearing and outside of diameter of gear shaft</td><td>SAR(4)-180</td><td rowspan="2">0.06 – 0.141</td><td colspan="2" rowspan="2">0.20</td></tr> <tr> <td></td><td></td><td>SAR(4)-180</td></tr> <tr> <td></td><td></td><td>SAR(1)-6</td><td>0.06 – 0.119</td><td>0.20</td></tr> <tr> <td>3</td><td>Depth to knock in pin</td><td>Model</td><td>Standard size</td><td>Tolerance</td><td>Repair limit</td></tr> <tr> <td></td><td></td><td>SAR(4)-180</td><td>21</td><td>0 –0.5</td><td rowspan="4">—</td></tr> <tr> <td></td><td></td><td>SAR(4)-180/ SAR(4)-180</td><td>14</td><td>0 –0.5</td></tr> <tr> <td></td><td></td><td>SAR(4)-180/ SAR(1)-6</td><td>14</td><td>0 –0.5</td></tr> <tr> <td></td><td></td><td>SAR(1)-6</td><td>10</td><td>0 –0.5</td></tr> <tr> <td>4</td><td>Rotating torque of spline shaft</td><td colspan="3">15.8 – 28.5 Nm {1.6 – 2.9 kgm}</td><td></td></tr> </tbody> </table>	No.	Check item	Criteria			Remedy			Model	Standard clearance	Clearance limit		1	Side clearance	SAR(4)-180	0.11 – 0.16	0.20	Replace			SAR(4)-180			SAR(1)-6	0.10 – 0.15	0.19	2	Clearance between inside diameter of plain bearing and outside of diameter of gear shaft	SAR(4)-180	0.06 – 0.141	0.20				SAR(4)-180			SAR(1)-6	0.06 – 0.119	0.20	3	Depth to knock in pin	Model	Standard size	Tolerance	Repair limit			SAR(4)-180	21	0 –0.5	—			SAR(4)-180/ SAR(4)-180	14	0 –0.5			SAR(4)-180/ SAR(1)-6	14	0 –0.5			SAR(1)-6	10	0 –0.5	4	Rotating torque of spline shaft	15.8 – 28.5 Nm {1.6 – 2.9 kgm}			
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NOTE

INSPECTION BY

APPROVE BY

MECHANIC

GROUP LEADER (GL)

REV. 00



PLANT REBUILD CENTER

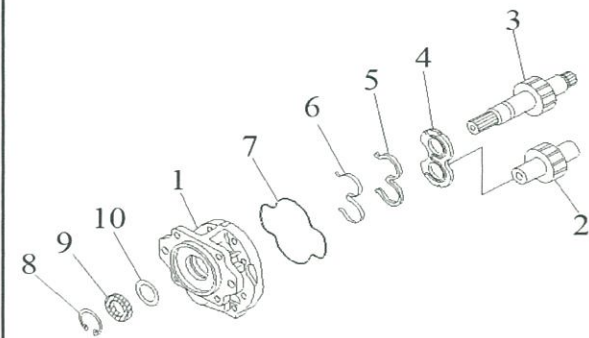
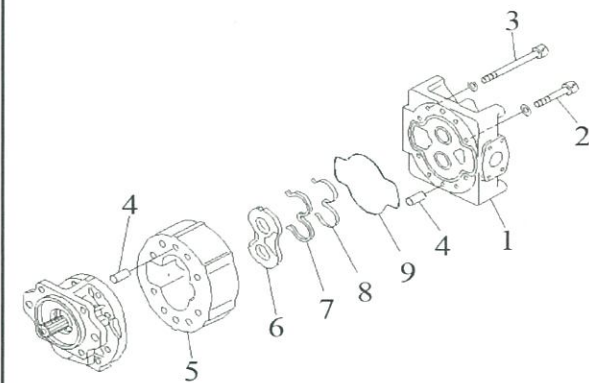
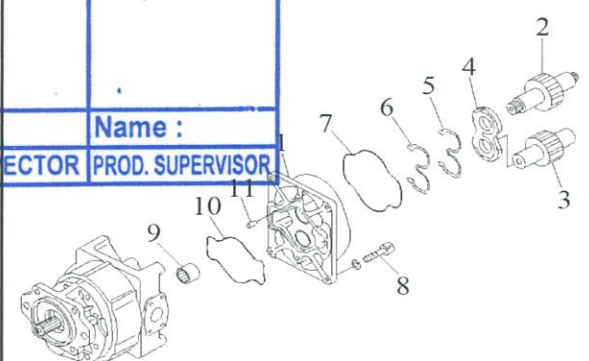
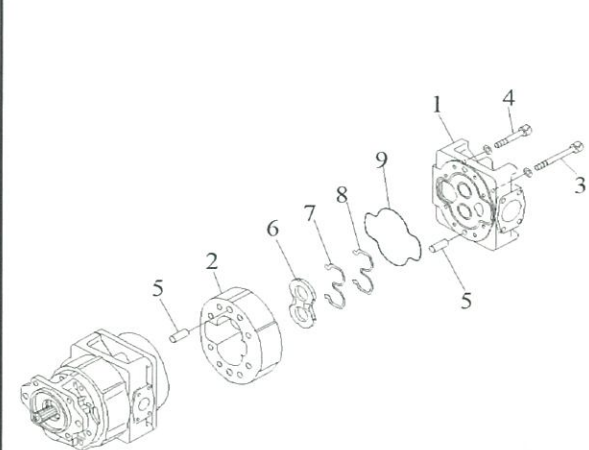
GEAR PUMP
SAR(4)180+180+(1)6
ASSEMBLY
CHECK SHEET

MACHINE MODEL

HD785-7

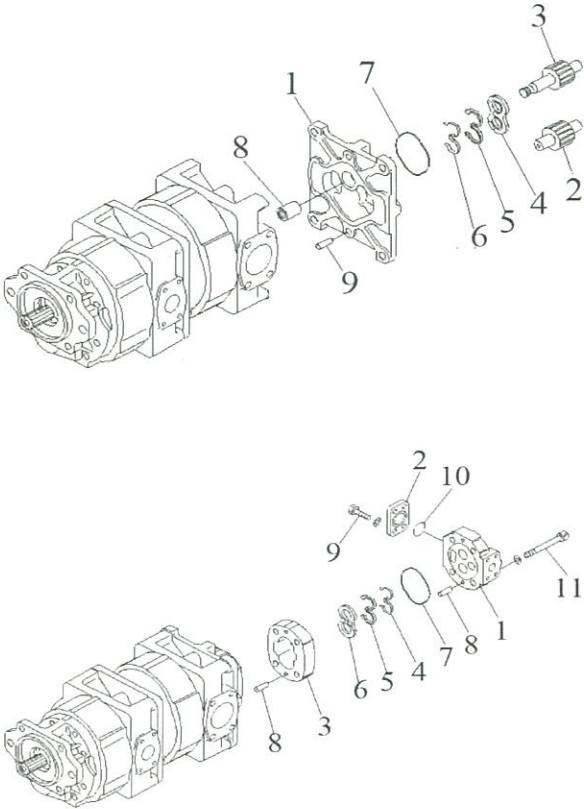
WO NO:

POWER TRAIN SECTION

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO												
GEAR PUMP		COMP.MODEL	PUBLIC DATE												
SAR(4)180+180+(1)6		COMP.S/N	REVISION NO												
		UNIT MODEL	PAGE												
PROCESS	START FINISH	ASSEMBLY BY													
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH GL/QA SKETCH DRAWING OR PHOTHOES												
1	SAR (4) 180 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 - Install plate (10) oil seal (9) and snapping (8) to bracket (1) - Install bhusing to bracket (1) - Install O-ring (7) to bracket (1) - install seal (6) and ring back up(5) to side plate (4) than install to bracket(1) - Install gear (3) and (2) to bracket (1) - Install pin dowel (4) to cradle(5) than Install them to bracket (1) - Install seal (7) and ring back up (8) to side plate (6) than install them to cradle (5) - Install bhusing to housing (1) - Install pindowel (4) and O-ring (9) to housing (1) than install them to cradle (5) - Install mounting bolt (2) and (3)  Tightening bolt (2)(3) <table border="1"> <tr> <td>Standart</td> <td>Actual</td> </tr> <tr> <td>18 ± 2.0 Kgm</td> <td></td> </tr> </table>	Standart	Actual	18 ± 2.0 Kgm		 								
Standart	Actual														
18 ± 2.0 Kgm															
2	SAR 180 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 - Install coupling (9) to housing - Install O-ring (10) pin dowel (11) and bushing to bracket (1) - Install bracket (1) assy to housing than Install bolt (8)  Tightening bolt (8) <table border="1"> <tr> <td>Standart</td> <td>Actual</td> </tr> <tr> <td>18 ± 2.0 Kgm</td> <td></td> </tr> </table> - Install O-ring (7) to bracket (1) - Install seal (5) and ring back up (6) to side plate (4) a than Install them to bracket (1) - Install gear (2) (3) to bracket (1) - fit pin dowel (5) to cradel (2) - Install cradle assy to housing - Install seal (7) and ring back up (8) to side plate (6) a than Install them to cradle (2) - Install O-ring (9) and dowel pin (5) to housing (1) a than Install them to cradle assy - Install bolt (3) (4) to housing (1)  Tightening bolt (3) (4) <table border="1"> <tr> <td>Standart</td> <td>Actual</td> </tr> <tr> <td>18 ± 2.0 Kgm</td> <td></td> </tr> </table>	Standart	Actual	18 ± 2.0 Kgm		Standart	Actual	18 ± 2.0 Kgm		RANDOM ITEM CHECK <table border="1"> <tr> <td>Name :</td> <td>Name :</td> </tr> <tr> <td>QA INSPECTOR</td> <td>PROD. SUPERVISOR</td> </tr> </table>  	Name :	Name :	QA INSPECTOR	PROD. SUPERVISOR
Standart	Actual														
18 ± 2.0 Kgm															
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ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO	
GEAR PUMP		COMP.MODEL	PUBLIC DATE	
SAR(4)180+180+(1)6		COMP.S/N	REVISION NO	
		UNIT MODEL	PAGE	

PROCESS	START FINISH	ASSEMBLY BY	
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NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES								
	SAR (1) 6 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 • Install coupling (8) to housing assy • Install pin dowel (9) and O-ring (7) to bracket (1) • Install seal (5) and ring back up (6) to side plate (4) a than Install them to bracket (1) • Install gear (2) (3) to bracket (1) • Install pin dowel (8) to cradle (3) a than Install them to housing assy • Install seal (5) and ring back up(4) to side plate (6) a than Install them to cradle (3) • Install O-ring (7) (2) to bracket (1) and Install pin dowel (8) than Install plate (2) to bracket (1) • Install mounting bolt (9) and (11)  Tightening bolt (9) <table border="1" style="width: 100%;"> <tr><td>Standart</td><td>Actual</td></tr> <tr><td>11.0 ± 1.0 Kgm</td><td></td></tr> </table>  Tightening bolt (11) <table border="1" style="width: 100%;"> <tr><td>Standart</td><td>Actual</td></tr> <tr><td>11 ± 1.0 Kgm</td><td></td></tr> </table>	Standart	Actual	11.0 ± 1.0 Kgm		Standart	Actual	11 ± 1.0 Kgm				
Standart	Actual												
11.0 ± 1.0 Kgm													
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NOTE

INSPECTION BY

MECHANIC

APPROVE BY

GROUP LEADER (GL)

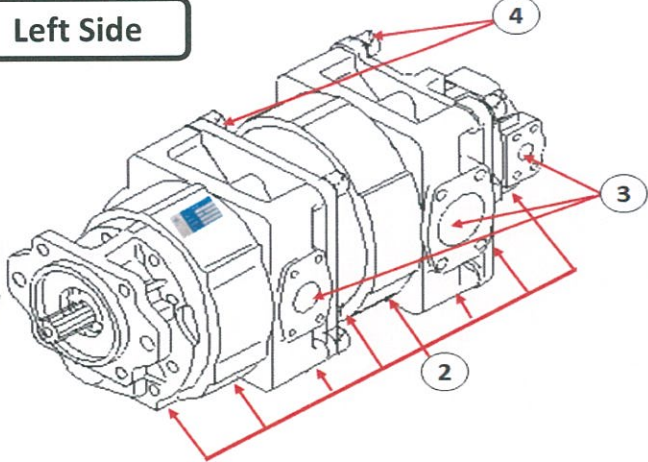
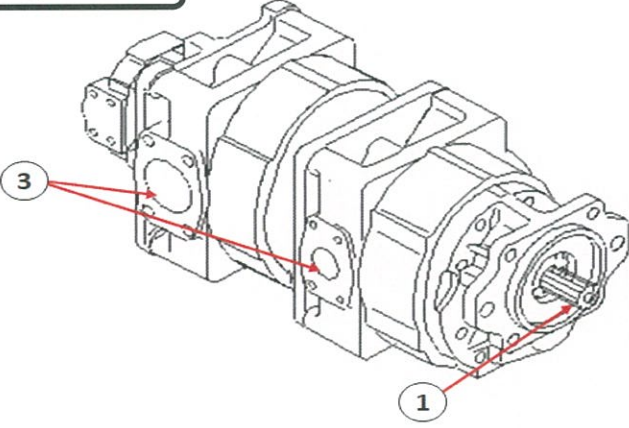
DELIVERY CHECK SHEET GEAR PUMP SAR (4) 180 + 180 + (1)6	WO. NO		DOCUMENT NO	
	COMP.S/N		PUBLIC DATE	
	INSPECTION DATE		REVISION NO	
	PART NO. COMPONENT	705 - 56 - 44090	PAGE	

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM	CHECK	REMARK
Left Side 	1	Shaft protection		
	2	Main housing		
	3	Hole protection		
	4	Bolt		
		Name plate		
		Painting condition		
		Packing / wraping		
		Stand		
		Bolt stand		
		Delivery check sheet		
		Quality passed (Sticker)		
	Right Side 			
NOTE : PLEASE MARK "✓" FOLLOWING TO THE ACTUAL CONDITION				
OIL LEVEL	EMPTY OR DRAINED			
	PROPER LEVEL			

Note :

INSPECTION BY

APPROVED BY

QA/GL

QA COORDINATOR

PLANT REBUILD CENTER

GEAR PUMP

SAR(4)180+180+(1)6

DELIVERY

CHECK SHEET

MACHINE MODEL

HD 785-7

WO NO:

POWER TRAIN SECTION

REV. 00



PLANT REBUILD CENTER

GEAR PUMP
SAR(30)100+100+SA(1)25
RECEIVING CHECK SHEET

MACHINE MODEL

HD785-7

WO NO:

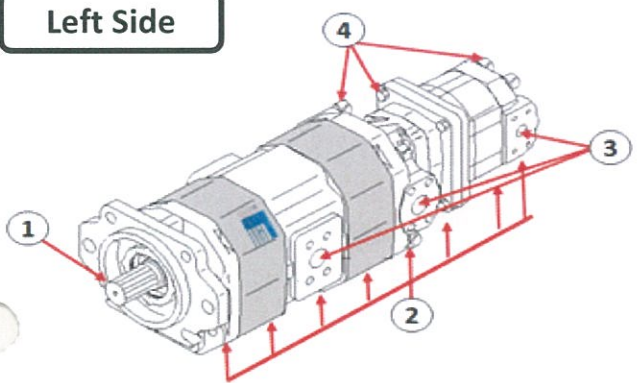
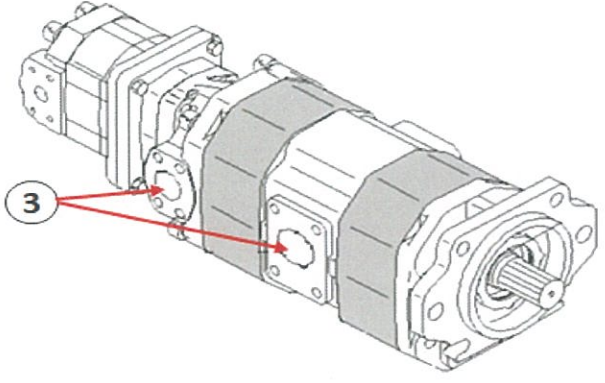
POWER TRAIN SECTION

RECEIVING CHECK SHEET GEAR PUMP SDR (30) 100+100+SA	WO. NO		DOCUMENT NO	
	COMP.S/N		PUBLIC DATE	
	INSPECTION DATE		REVISION NO	
	JOB SITE		PAGE	

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM	CHECK	REMARK						
Left Side 	1	Spline of shaft								
	2	Main housing								
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	4	Mounting bolt								
		Name plate								
		Painting condition								
		Packing / wraping								
		Stand								
		Bolt stand								
Right Side 	3									
NOTE : PLEASE MARK "✓" FOLLOWING TO THE ACTUAL CONDITION										
<table border="1"> <tr> <td>OIL LEVEL</td> <td>EMPTY OR DRAINED</td> <td></td> </tr> <tr> <td></td> <td>PROPER LEVEL</td> <td></td> </tr> </table>		OIL LEVEL	EMPTY OR DRAINED			PROPER LEVEL				
OIL LEVEL	EMPTY OR DRAINED									
	PROPER LEVEL									

Note :

INSPECTION BY

CHECK BY

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QA COORDINATOR

GL/SPV PROD

PPC

REV. 00



PLANT REBUILD CENTER

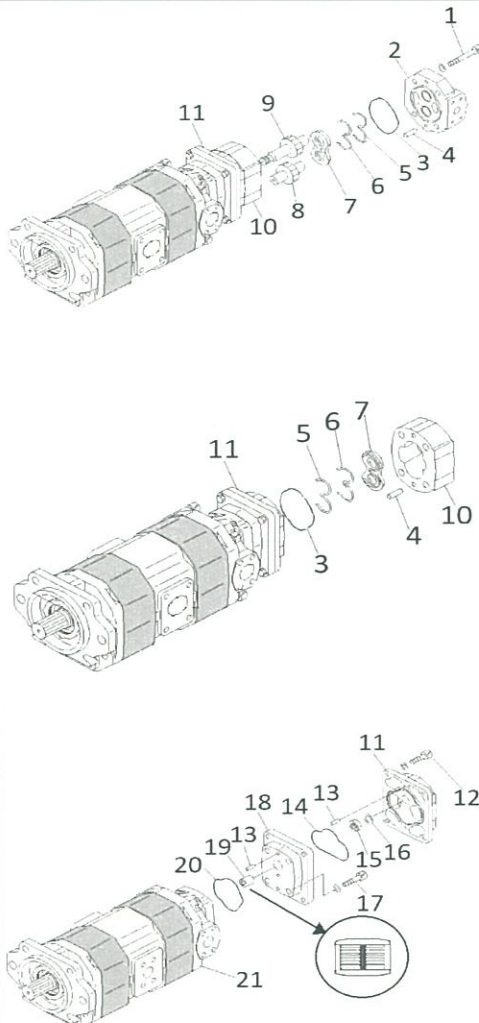
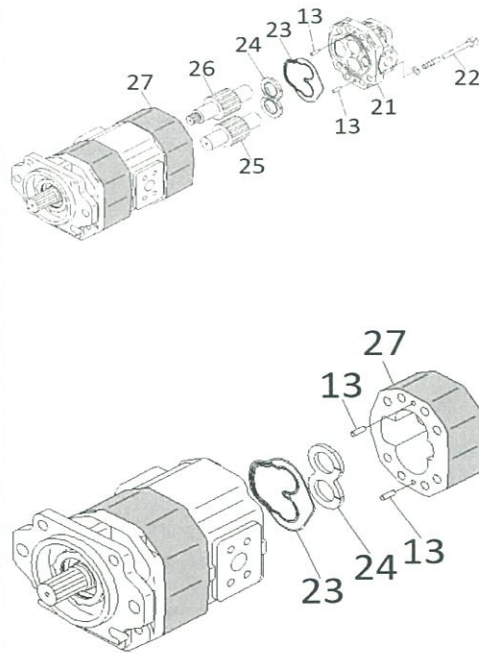
GEAR PUMP
SAR(30)100+100+SA(1)25
DISSASSEMBLY CHECK SHEET

MACHINE MODEL

HD785-7

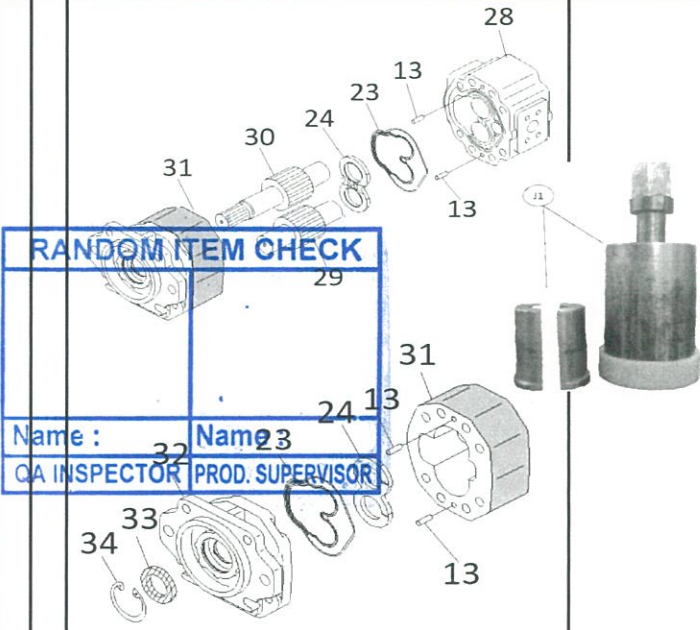
WO NO:

POWER TRAIN SECTION

DISASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO		
GEAR PUMP		COMP.MODEL	PUBLIC DATE		
SDR(30)100+100+SA(1)25		COMP.S/N	REVISION NO		
		UNIT MODEL	PAGE		
PROCESS		START	DISASSEMBLY BY		
		FINISH			
NO	PART TO REMOVE OR INSPECT	CONDITIONS TOBE INSPECTION	REUSE SALVAGE REPAIR	SKETCH DRAWING OR PHOTOOES	REMARK
1	SA(1)25 DISASSEMBLY	⚠ IMPORTANT ⚠ Check contact surface cover and bracket,treath,and abnormal worn of pin dowel.Check abnormal worn of gear and side plate contact surface gear don't mix • Remove bolt (1)from bracket (11). •Tracking out housing (4) by hammer slowly. • Check dowel pin & holes • Remove O-ring (3) from cover (2) • Remove side plate assy (7) ring back up (5) and seal oil (6) • Remove gear (8) and gear (9) • Remove cradel (10) by bloe hammer • Remoeve pin dowel (4) from cradel (10) • Remove side plate assy (7) ring back up (5) and seal oil (6) • Remove O-ring (3) from bracket (11) • Remove bolt (12) from bracket (18) • Remoeve bracket (11) from bracket (18) • Remove pin dowel (13) plate (16) seal oil (15) and O-ring (14) from bracket (11) • Remove bolt (17) from bracket (21) • Remove bracket (18) from bracket (21) • Remove pin dowel (13) coupling (19) and O-ring (20) from bracket (18)			
2	SDR(30)100 DISASSEMBLY	⚠ IMPORTANT ⚠ Check contact surface cover and bracket,treath,and abnormal worn of pin dowel.Check abnormal worn of gear and side plate contact surface gear don't mix • Remove bolt (12) • Remove bracket (21) • Remove pin dowel (13) seal ring (23) • Remove side plate (24) and gear (25)(26) • Remove cradle (27) • Remove pin dowel (13) • Remove side plate (24) and seal ring (23)			

DISASSEMBLY CHECK SHEET GEAR PUMP SDR(30)100+100+SA(1)25	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	R 00
	UNIT MODEL		PAGE	3 of 6

PROCESS	START		DISASSEMBLY BY	
	FINISH			

NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	REUSE	SALVAGE	REPAIR	SKETCH DRAWING OR PHOTOES	REMARK
3	SDR(30)100 DISASSEMBLY	⚠ IMPORTANT ⚠ Check contact surface cover and bracket,treath,and abnormal worn of pin dowel.Check abnormal worn of gear and side plate contact surface gear don't mix - Remove carrier (28) and pin dowel (13) - Remove seal ring (23) and side plate (24) - Remove gear (29) and (30) - Remove cradle (31) and pin dowel (13) - Remove side plate (24) and seal ring (23) - Remove oil seal (33) and snap ring (34) - Remove all bhusing with tool J1					

NOTE

INSPECTION BY

APPROVE BY

MECHANIC

GROUP LEADER (GL)

REV. 00



PLANT REBUILD CENTER

GEAR PUMP
SAR(30)100+100+SA(1)25
MEASSUREMENT
CHECK SHEET

MACHINE MODEL

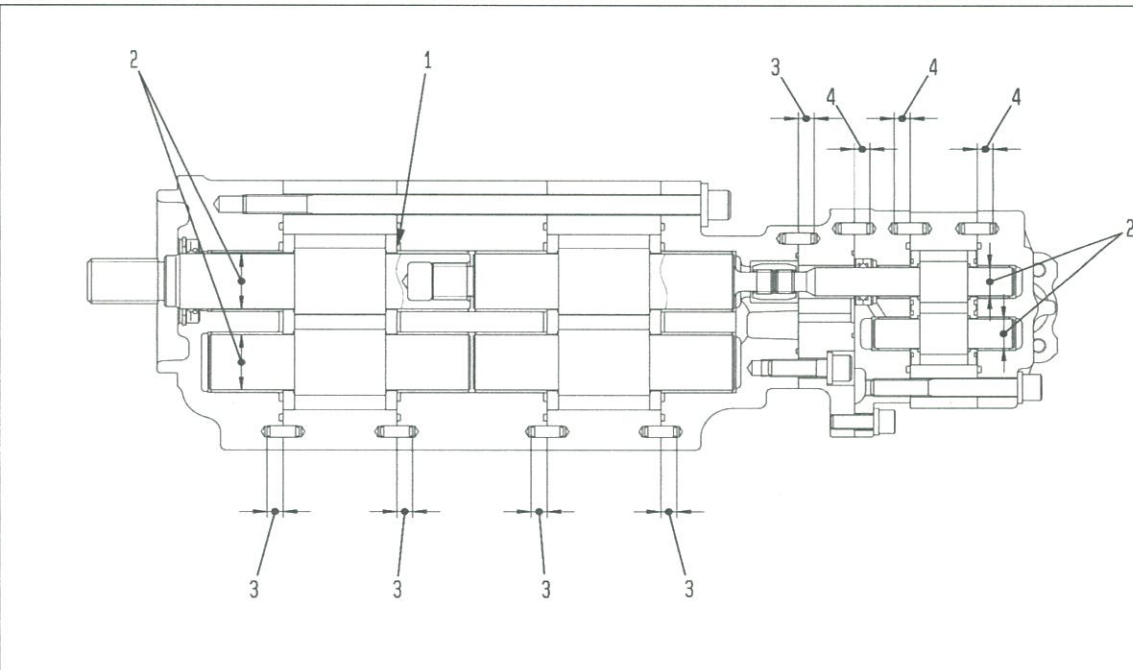
HD785-7

WO NO:

POWER TRAIN SECTION

MEASUREMENT CHECK SHEET	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	R 00
	SDR(30)100+100+SA(1)25	UNIT MODEL	PAGE	4 of 6

PROCESS	START		MEASUREMENT BY		
	FINISH				

NO	PART TO MEASURING OR INSPECT	STANDARD VALUE TABLE																																																
1	SDR(30)100+100+SA(1)25 MEASUREMENT																																																	
		Unit: mm <table border="1"> <thead> <tr> <th data-bbox="363 1099 408 1131">No.</th><th data-bbox="408 1099 718 1131">Check item</th><th colspan="2" data-bbox="718 1099 933 1131">Criteria</th><th data-bbox="933 1099 1501 1131"></th></tr> <tr> <th data-bbox="363 1131 408 1160"></th><th data-bbox="408 1131 718 1160"></th><th data-bbox="718 1131 933 1160">Model</th><th data-bbox="933 1131 1139 1160">Standard clearance</th><th data-bbox="1139 1131 1501 1160">Clearance limit</th></tr> </thead> <tbody> <tr> <td data-bbox="363 1160 408 1227">1</td><td data-bbox="408 1160 718 1227" rowspan="2">Side clearance</td><td data-bbox="718 1160 933 1191">SDR(30)-100</td><td data-bbox="933 1160 1139 1191">0.16 – 0.20</td><td data-bbox="1139 1160 1501 1191">0.24</td></tr> <tr> <td data-bbox="363 1191 408 1227"></td><td data-bbox="718 1191 933 1227">SA(1)-25</td><td data-bbox="933 1191 1139 1227">0.10 – 0.15</td><td data-bbox="1139 1191 1501 1227">0.19</td></tr> <tr> <td data-bbox="363 1227 408 1323">2</td><td data-bbox="408 1227 718 1323" rowspan="2">Clearance between inside diameter of plain bearing and outside of diameter of gear shaft</td><td data-bbox="718 1227 933 1258">SDR(30)-100</td><td data-bbox="933 1227 1139 1258">0.06 – 0.131</td><td data-bbox="1139 1227 1501 1258" rowspan="2">0.20</td></tr> <tr> <td data-bbox="363 1258 408 1323"></td><td data-bbox="718 1258 933 1323">SA(1)-25</td><td data-bbox="933 1258 1139 1323">0.06 – 0.119</td></tr> <tr> <td data-bbox="363 1323 408 1355">3</td><td data-bbox="408 1323 718 1355" rowspan="2">Depth to knock in pin</td><td data-bbox="718 1323 933 1355">Model</td><td data-bbox="933 1323 1139 1355">Standard size</td><td data-bbox="1139 1323 1501 1355">Tolerance</td></tr> <tr> <td data-bbox="363 1355 408 1386">4</td><td data-bbox="718 1355 933 1386">SDR(30)-100</td><td data-bbox="933 1355 1139 1386" rowspan="3">10</td><td data-bbox="1139 1355 1501 1386" rowspan="3">—</td></tr> <tr> <td data-bbox="363 1386 408 1415"></td><td></td><td data-bbox="718 1386 933 1415">SA(1)-25</td></tr> <tr> <td data-bbox="363 1415 408 1458">5</td><td data-bbox="408 1415 718 1458">Rotating torque of spline shaft</td><td colspan="3" data-bbox="718 1415 1501 1458">8.9 – 16.7 Nm {0.9 – 1.7 kgm}</td></tr> </tbody> </table>					No.	Check item	Criteria					Model	Standard clearance	Clearance limit	1	Side clearance	SDR(30)-100	0.16 – 0.20	0.24		SA(1)-25	0.10 – 0.15	0.19	2	Clearance between inside diameter of plain bearing and outside of diameter of gear shaft	SDR(30)-100	0.06 – 0.131	0.20		SA(1)-25	0.06 – 0.119	3	Depth to knock in pin	Model	Standard size	Tolerance	4	SDR(30)-100	10	—			SA(1)-25	5	Rotating torque of spline shaft	8.9 – 16.7 Nm {0.9 – 1.7 kgm}		
No.	Check item	Criteria																																																
		Model	Standard clearance	Clearance limit																																														
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		SA(1)-25	0.06 – 0.119																																															
3	Depth to knock in pin	Model	Standard size	Tolerance																																														
4		SDR(30)-100	10	—																																														
		SA(1)-25																																																
5	Rotating torque of spline shaft	8.9 – 16.7 Nm {0.9 – 1.7 kgm}																																																

[illegible]

REV. 00



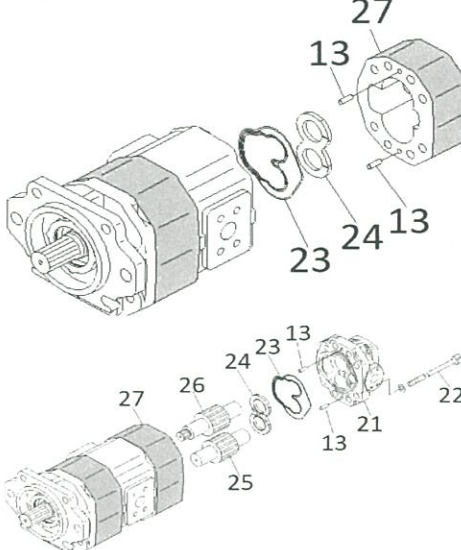
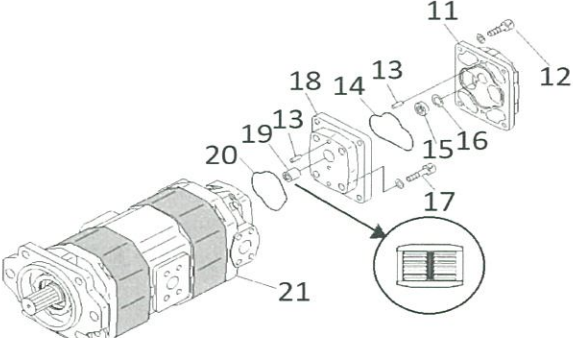
PLANT REBUILD CENTER

GEAR PUMP
SAR(30)100+100+SA(1)25
ASSEMBLY
CHECK SHEET

MACHINE MODEL

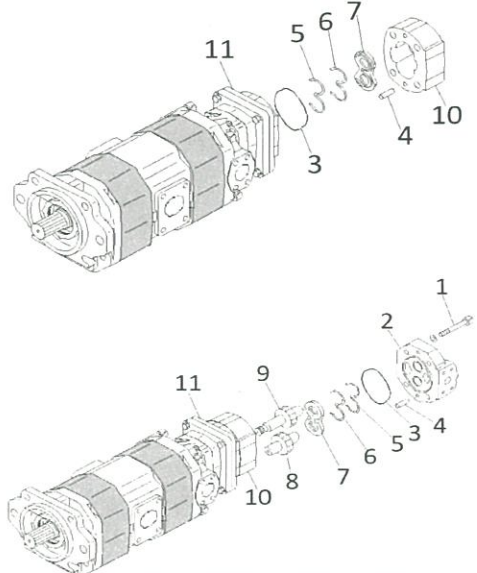
HD785-7

WO NO:

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO								
GEAR PUMP		COMP.MODEL	PUBLIC DATE								
SDR(30)100+100+SA(1)25		COMP.S/N	REVISION NO								
		UNIT MODEL	PAGE								
			R 00								
			5 of 6								
PROCESS	START FINISH	ASSEMBLY BY									
NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH GL/QA SKETCH DRAWING OR PHOTHOES								
1	SDR(30)100 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 - Install oil seal (33) and snap ring (34) to bracket (32) - Install bushing to bracket (32) - Fit seal ring (23) to bracket (32) - Fit side plate (24) to bracket (32) - Install pin dowel (13) to cradle (31) a than install them to bracket (32) - Install gear (30) (29) and install side plate (24) to cradle - Install seal ring (23) to bracket (28) a than install them to cradle (31) - Install all bhushing with tool J2	 GA INSPECTOR PROD. SUPERVISOR								
2	SDR(30)100 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 - Install ring seal (23) and side plate (24) to bracket (28) - Intal pin dowel (13) to cradle (27) a than install them to bracket (28) - Install side plate (24) and seal ring (23) to bracket(21) - Install pindowel (13) to bracet (21) athan Install them to cradle (27)  Tightening bolt (22) <table border="1"> <tr> <th>Standart</th> <th>Actual</th> </tr> <tr> <td>18 ± 2.5 Kgm</td> <td></td> </tr> </table>	Standart	Actual	18 ± 2.5 Kgm						
Standart	Actual										
18 ± 2.5 Kgm											
3	SA(1)25 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 - Install O-ring (20) to bracket (19) - Install coupling (17) pin dowel (13) to bracket(18) a than Install them to bracket (21)  Tightening bolt (17) <table border="1"> <tr> <th>Standart</th> <th>Actual</th> </tr> <tr> <td>10 ~ 12.5 Kgm</td> <td></td> </tr> </table> - Install O-ring (14) to cover (18) - Install pin dowel (13) plate (16)and seal oil (15) to Bracket (11)  Tightening bolt (12) <table border="1"> <tr> <th>Standart</th> <th>Actual</th> </tr> <tr> <td>6 ~ 7.5 Kgm</td> <td></td> </tr> </table>	Standart	Actual	10 ~ 12.5 Kgm		Standart	Actual	6 ~ 7.5 Kgm		
Standart	Actual										
10 ~ 12.5 Kgm											
Standart	Actual										
6 ~ 7.5 Kgm											

ASSEMBLY CHECK SHEET		WO NUMBER		DOCUMENT NO	
GEAR PUMP		COMP.MODEL		PUBLIC DATE	
SDR(30)100+100+SA(1)25		COMP.S/N		REVISION NO	
		UNIT MODEL		PAGE	
				R 00	
				6 of 6	

PROCESS	START		ASSEMBLY BY		
	FINISH				

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES				
3	SA(1)25 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 Install O-ring (3) side plate (7) seal (6) and back up (5) Install pin dowel (4) to cradle (10) Install gear (8) and gear (9) to cradle (10) Install side plate (7) back up(5) and seal (6) to cover (2)  Tightening bolt (1) <table border="1" style="margin-left: 20px;"> <tr> <td style="text-align: center;">Standart</td> <td style="text-align: center;">Actual</td> </tr> <tr> <td style="text-align: center;">10 ~ 12.5 Kgm</td> <td></td> </tr> </table>	Standart	Actual	10 ~ 12.5 Kgm				
Standart	Actual								
10 ~ 12.5 Kgm									

NOTE

INSPECTION BY

APPROVE BY

MECHANIC

GROUP LEADER (GL)

PLANT REBUILD CENTER

GEAR PUMP

SAR(30)100+100+SA(1)25

DELIVERY

CHECK SHEET

MACHINE MODEL

HD 785-7

WO NO:

POWER TRAIN SECTION

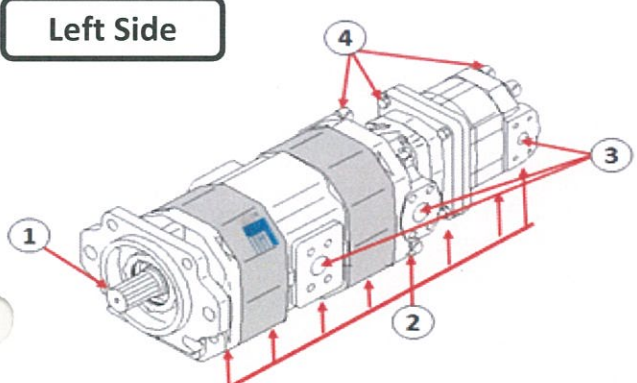
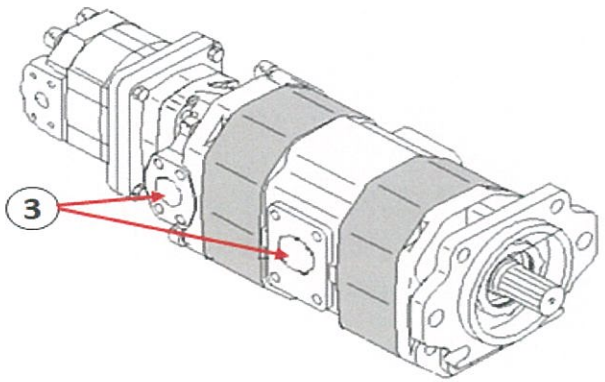
DELIVERY CHECK SHEET GEAR PUMP SDR (30) 100+100+SA	WO. NO		DOCUMENT NO	
	COMP.S/N		PUBLIC DATE	
	INSPECTION DATE		REVISION NO	
	PART NO. COMPONENT	705 - 95 - 07121	PAGE	

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM	CHECK	REMARK
Left Side 	1	Shaft protection		
	2	Main housing		
	3	Hole protection		
	4	Bolt		
		Name plate		
		Painting condition		
		Packing / wrapping		
		Stand		
		Bolt stand		
		Delivery check sheet		
		Quality passed (Sticker)		
Right Side 				
NOTE : PLEASE MARK "✓" FOLLOWING TO THE ACTUAL CONDITION				
OIL LEVEL	EMPTY OR DRAINED			
	PROPER LEVEL			

Note :

INSPECTION BY

APPROVED BY

QA/GL

QA COORDINATOR

REV. 00



PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

GEAR PUMP SAR(4)180+100

RECEIVING

CHECK SHEET

MACHINE MODEL

HD785-7

WO NO:

POWER TRAIN SECTION

RECIIVING CHECK SHEET

GEAR PUMP

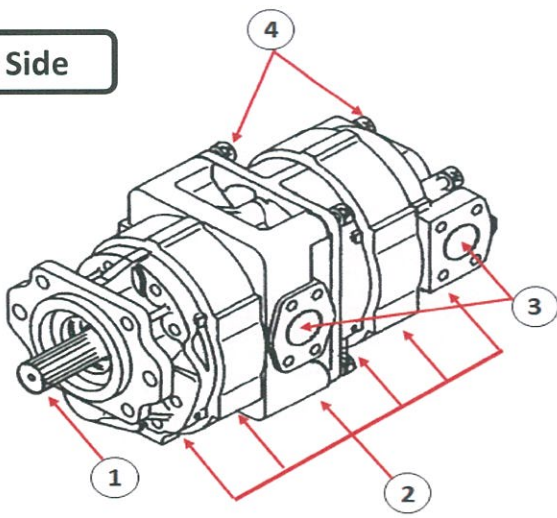
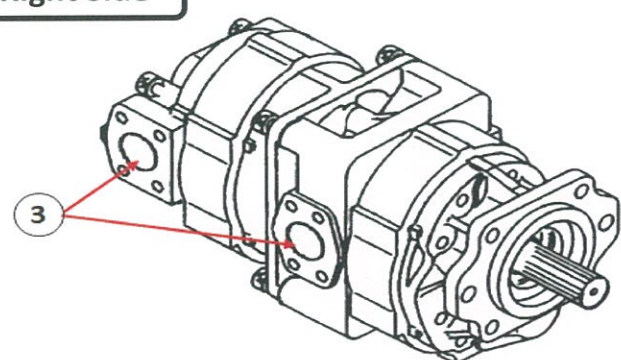
SAR (4) 180 + 180

WO. NO		DOCUMENT NO	
COMP.S/N		PUBLIC DATE	
INSPECTION DATE		REVISION NO	
JOB SITE		PAGE	

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM	CHECK	REMARK
Left Side 	1	Spline of shaft		
	2	Main housing		
	3	contact surface O-ring and treath (Discharge)(Suction)		
	4	Mounting bolt		
		Name plate		
		Painting condition		
		Packing / wraping		
		Stand		
		Bolt stand		
Right Side 				
NOTE : PLEASE MARK "✓" FOLLOWING TO THE ACTUAL CONDITION				
OIL LEVEL	EMPTY OR DRAINED			
	PROPER LEVEL			

Note :

INSPECTION BY

CHECK BY

CHECK BY

APPROVED BY

QA INSPECTOR

QA COORDINATOR

GL/SPV PROD

PPC

REV. 00



PLANT REBUILD CENTER

GEAR PUMP SAR(4)180+100
DISSASSEMBLY
CHECK SHEET

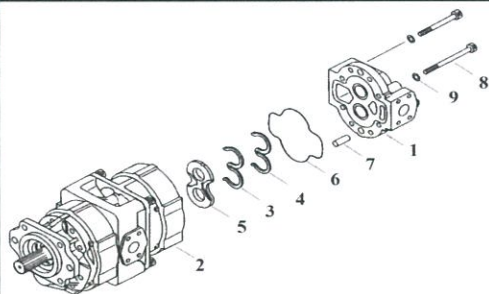
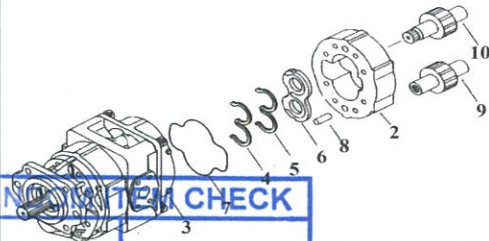
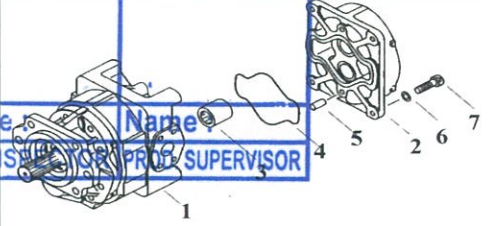
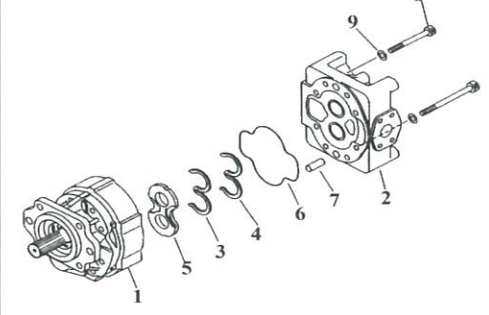
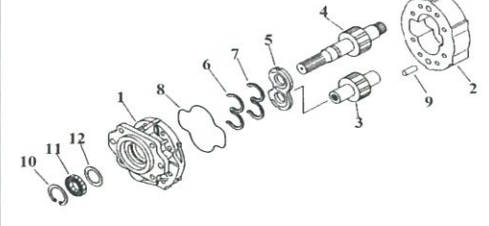
MACHINE MODEL

HD785-7

WO NO:

POWER TRAIN SECTION

DISASSEMBLY CHECK SHEET GEAR PUMP SAR (4)180+100		WO NUMBER		DOCUMENT NO	
		COMP.MODEL		PUBLIC DATE	
		COMP.S/N		REVISION NO	
		UNIT MODEL		PAGE	
PROCESS	START FINISH		DISASSEMBLY BY		

NO	PART TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTION	REUSE	SALVAGE	REPAIR	SKETCH DRAWING OR PHOTOES	REMARK
1	SAR (4)180 DISASSEMBLY	⚠ IMPORTANT ⚠ Check contact surface cover and bracket,treath,and abnormal worn of pin dowel.Check abnormal worn of gear and side plate contact surface gear don't mix • Reove bolt (8) from bracket (1) • Tracking bracket (1) by blow hammer • Remove dowel pin (7) O-ring (6) from bracket (1) • Remove bushing from bracket (1) by bushing puller • Remove side plate (5)seal(4) and ring back up(3) from cradle (2) • Remove gear (9) and (10) from cradle (2) • Remove cradle (2) from bracket (3) • Remove pin dowel (8) from cradle (2) • Remove side plate (6) seal (4) ring back up (5) from cradle (2) • Remove O-ring (7) from bracket (3)				 	
2	SAR 100 DISASSEMBLY	⚠ IMPORTANT ⚠ Check contact surface cover and bracket,treath,and abnormal worn of pin dowel.Check abnormal worn of gear and side plate contact surface gear don't mix • Remove bolt (7) from bracket (2) • Tracking bracket (2) by blow hammer • Remove pin dowel(5) and O-ring (4) from bracket (2) • Remove coupling (3) from housing (1) • Remove bolt (8) from bracket (2) • Tracking bracket (2) by blow hammer • Remove pin dowel (7) O-ring(6) from bracket (2) a than remove bhusing by bushing puller • Remove side plate assy (5) • Remove cradle (2) by blow hammer • Remove gear (3)(4) from bracket (1) • Remove side plate assy (5) • Remove O-ring (8) plate(12) oil seal (11) and snap ring (10) from bracket (1)				  	

NOTE

INSPECTION BY

APPROVE BY

MECHANIC

GROUP LEADER (GL)

REV. 00



PLANT REBUILD CENTER

GEAR PUMP SAR(4)180+100

MEASUREMENT

CHECK SHEET

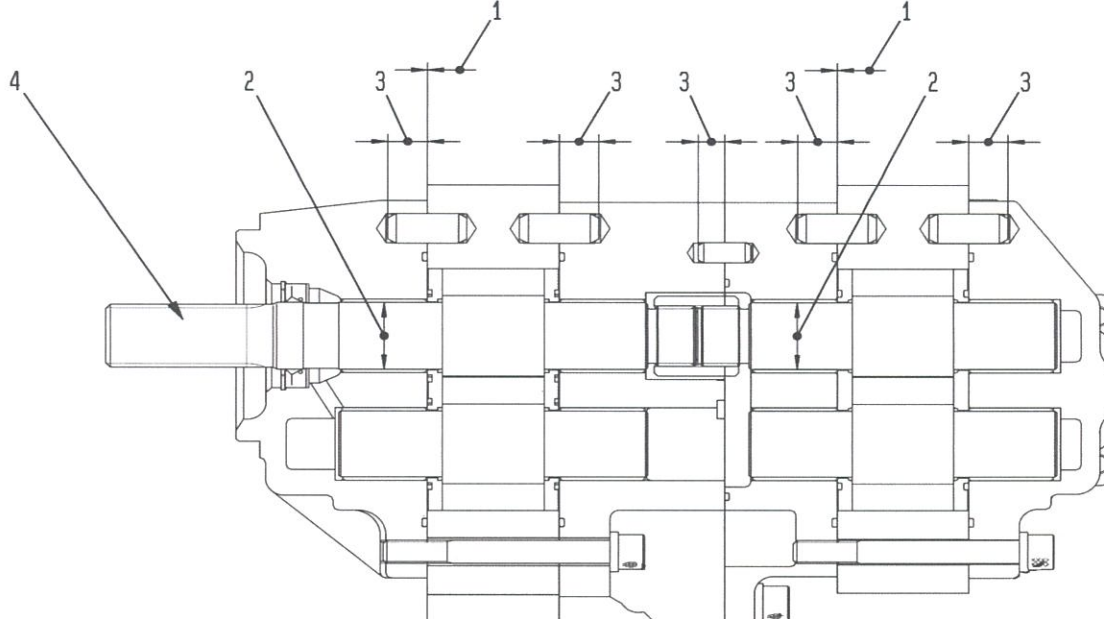
MACHINE MODEL

HD785-7

WO NO:

POWER TRAIN SECTION

MEASUREMENT CHECK SHEET GEAR PUMP SAR (4)180+100		WO NUMBER		DOCUMENT NO	
		COMP.MODEL		PUBLIC DATE	
		COMP.S/N		REVISION NO	
		UNIT MODEL		PAGE	
PROCESS	START FINISH		MEASUREMENT BY		

NO	PART TO MEASURING OR INSPECT	STANDARD VALUE TABLE																																											
1	MEASURING BREAK COOLING PUMP ASSEMBLY																																												
		Unit: mm <table border="1" data-bbox="375 1039 1513 1341"> <thead> <tr> <th data-bbox="375 1039 418 1072">No.</th><th data-bbox="418 1039 729 1072">Check item</th><th colspan="2" data-bbox="729 1039 892 1072">Criteria</th><th data-bbox="892 1039 1513 1072">Remedy</th></tr> </thead> <tbody> <tr> <td data-bbox="375 1072 418 1128" rowspan="2">1</td><td data-bbox="418 1072 729 1128" rowspan="2">Clearance between gear case and side plate, gear</td><td data-bbox="729 1072 892 1095">Model</td><td data-bbox="892 1072 1158 1095">Standard clearance</td><td data-bbox="1158 1072 1513 1095">Clearance limit</td></tr> <tr> <td data-bbox="729 1095 892 1128">SAR(4)-180</td><td data-bbox="892 1095 1158 1128">0.11 – 0.16</td><td data-bbox="1158 1095 1513 1128">0.20</td></tr> <tr> <td data-bbox="375 1128 418 1207">2</td><td data-bbox="418 1128 729 1207">Clearance between inside diameter of plain bearing and outside diameter of gear shaft</td><td data-bbox="729 1128 892 1207">SAR(4)-180</td><td data-bbox="892 1128 1158 1207">0.06 – 0.14</td><td data-bbox="1158 1128 1513 1207">0.19</td></tr> <tr> <td data-bbox="375 1207 418 1341" rowspan="3">3</td><td data-bbox="418 1207 729 1341" rowspan="3">Depth to knock in pin</td><td data-bbox="729 1207 892 1240">Model</td><td data-bbox="892 1207 1054 1240">Standard size</td><td data-bbox="1054 1207 1217 1240">Tolerance</td></tr> <tr> <td data-bbox="729 1240 892 1274">SAR(4)-180</td><td data-bbox="892 1240 1054 1274">21</td><td data-bbox="1054 1240 1217 1274">0</td></tr> <tr> <td data-bbox="729 1274 892 1341">SAR(4)-180/ SAR(4)-180</td><td data-bbox="892 1274 1054 1341">14</td><td data-bbox="1054 1274 1217 1341">-0.5</td></tr> <tr> <td data-bbox="375 1207 418 1341"></td><td data-bbox="418 1207 729 1341"></td><td data-bbox="729 1207 892 1341"></td><td data-bbox="892 1207 1054 1341"></td><td data-bbox="1054 1207 1217 1341">Repair limit</td></tr> <tr> <td data-bbox="375 1207 418 1341"></td><td data-bbox="418 1207 729 1341"></td><td data-bbox="729 1207 892 1341"></td><td data-bbox="892 1207 1054 1341"></td><td data-bbox="1054 1207 1513 1341">Replace</td></tr> </tbody> </table>					No.	Check item	Criteria		Remedy	1	Clearance between gear case and side plate, gear	Model	Standard clearance	Clearance limit	SAR(4)-180	0.11 – 0.16	0.20	2	Clearance between inside diameter of plain bearing and outside diameter of gear shaft	SAR(4)-180	0.06 – 0.14	0.19	3	Depth to knock in pin	Model	Standard size	Tolerance	SAR(4)-180	21	0	SAR(4)-180/ SAR(4)-180	14	-0.5					Repair limit					Replace
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		SAR(4)-180/ SAR(4)-180	14	-0.5																																									
				Repair limit																																									
				Replace																																									

NOTE

NOTE

INSPECTION BY

APPROVE BY

MECHANIC

GROUP LEADER (GL)

REV. 00



PLANT REBUILD CENTER

GEAR PUMP SAR(4)180+100

ASSEMBLY

CHECK SHEET

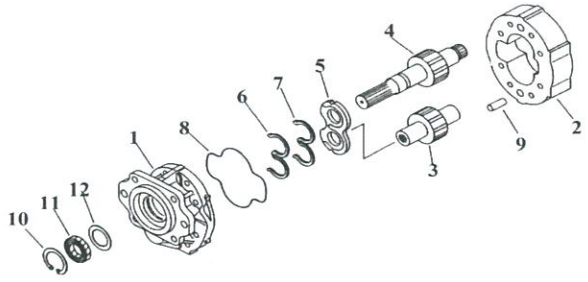
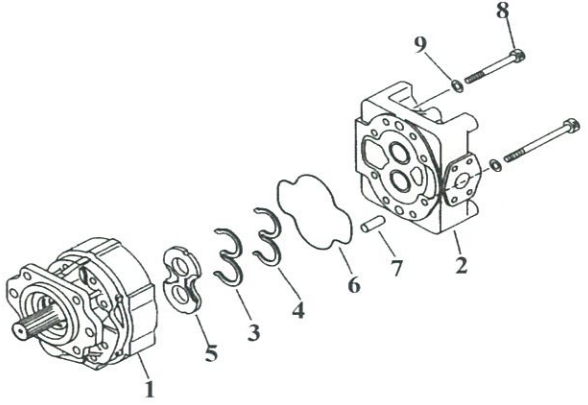
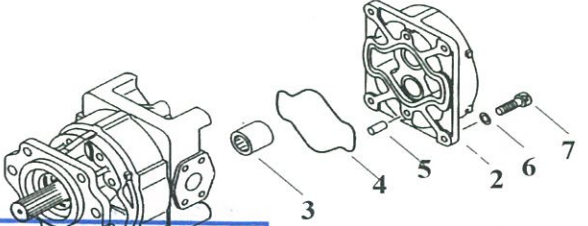
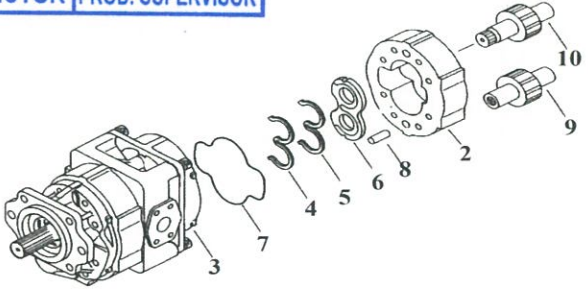
MACHINE MODEL

HD785-7

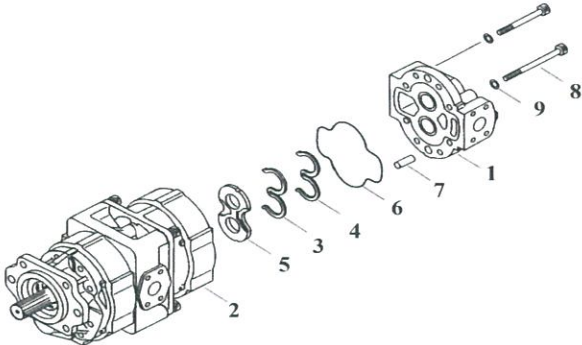
WO NO:

POWER TRAIN SECTION

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO
GEAR PUMP		COMP.MODEL	PUBLIC DATE
SAR (4)180+100		COMP.S/N	REVISION NO
		UNIT MODEL	PAGE
PROCESS	START FINISH	ASSEMBLY BY	

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TOBE INSTALL	MECH	GL/QA	SKETCH DRAWING OR PHOTHOES								
1	SAR (4) 180 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 Install Bhusing to bracket (1) Install plate (12) oil seal (11) and Install ring snap (10) to bracket (1) Install O-ring (8) to bracket (1) Install Oil seal (7) and ring back up (5) to side plate (5) a than Install them to bracket (1) Install gear (3) and (4) to bracket (1) Install pin dowel (9) to cradle (2) a than Install them to bracket (1) Install Oil seal (3) and ring back up (4) to side plate (5) a than Install them to cradle (1) Install O-ring (6) to bracket (2) Install pin dowel (7) to bracket (2) a than Install them to bracket (1) Install Bolt (8) to bracket (2)  Tightening bolt (8) <table><tr><th>Standart</th><th>Actual</th></tr><tr><td>18 ± 2.0 Kgm</td><td></td></tr></table> Install coupling (3) to bracket (1) Install O-ring (4) to bracket (2) Install pin dowel (5) to bracket (2) a than Install them to bracket (1) Install bolt (7) to bracket (2)  Tightening bolt (7) <table><tr><th>Standart</th><th>Actual</th></tr><tr><td>18 ± 2.0 Kgm</td><td></td></tr></table>	Standart	Actual	18 ± 2.0 Kgm		Standart	Actual	18 ± 2.0 Kgm				  
Standart	Actual												
18 ± 2.0 Kgm													
Standart	Actual												
18 ± 2.0 Kgm													
2	SAR 180 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 Install O-ring (7) to bracket (3) Install oil seal (5) and ring back up (4) to side plate (6) a than Install them to bracket (3) Install pin dowel (8) to cradle (2) a than Install them to bracket (3) Install gear (9) and gear (10) to cradle (2)		NAME : QA INSPECTOR NAME : PROD. SUPERVISOR									

ASSEMBLY CHECK SHEET GEAR PUMP SAR (4)180+100		WO NUMBER		DOCUMENT NO	
		COMP.MODEL		PUBLIC DATE	
		COMP.S/N		REVISION NO	
		UNIT MODEL		PAGE	
PROCESS	START FINISH		ASSEMBLY BY		

NO	PART TO INSTALL OR ASSEMBLY	CONDITION TO BE INSTALLED	MECH	GL/QA	SKETCH DRAWING OR PHOTOGRAPH				
2	SAR 180 ASSEMBLY	⚠ IMPORTANT ⚠ Inspect each part thoroughly and confirm that the part is free from dirt/dust, scratches.  Loctite 515 And engine oil SAE 30 Install Oil seal (3) and ring back up (4) to side plate (5) and then install them to cradle (2) Install O-ring (6) to bracket (1) Install pin dowel (7) to bracket (1) and then install them to cradle (2) Install bolt (8) to bracket (1)  Tightening bolt (7) <table border="1" data-bbox="397 779 544 835"> <tr><td>Standard</td></tr> <tr><td>18 ± 2.0 Kgm</td></tr> </table> <table border="1" data-bbox="611 779 758 835"> <tr><td>Actual</td></tr> <tr><td> </td></tr> </table>	Standard	18 ± 2.0 Kgm	Actual				
Standard									
18 ± 2.0 Kgm									
Actual									

[illegible]

PLANT REBUILD CENTER

GEAR PUMP

SAR(4)180+180

DELIVERY

CHECK SHEET

MACHINE MODEL

HD 785-7

WO NO:

POWER TRAIN SECTION

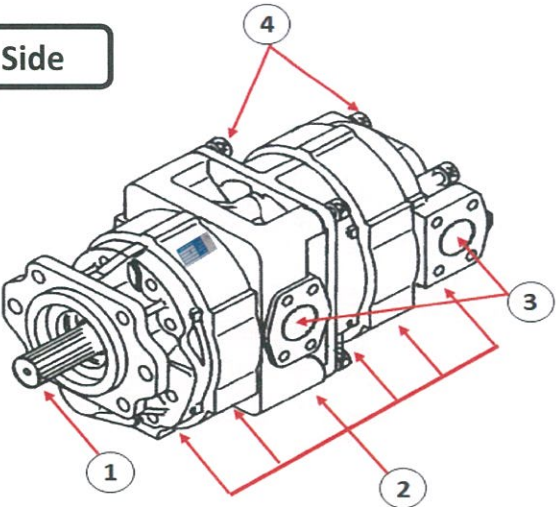
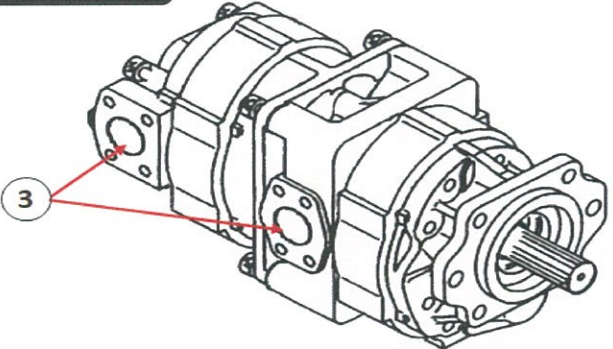
DELIVERY CHECK SHEET GEAR PUMP SAR (4) 180 + 180	WO. NO		DOCUMENT NO	
	COMP.S/N		PUBLIC DATE	
	INSPECTION DATE		REVISION NO	
	PART NO. COMPONENT	705 - 52 - 42220	PAGE	

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM	CHECK	REMARK
Left Side 	1	Shaft protection		
	2	Main housing		
	3	Hole protection		
	4	Bolt		
		Name plate		
		Painting condition		
		Packing / wrapping		
		Stand		
		Bolt stand		
		Delivery check sheet		
		Quality passed (Sticker)		
Right Side 				

NOTE : PLEASE MARK "✓" FOLLOWING TO THE ACTUAL CONDITION

OIL LEVEL	EMPTY OR DRAINED	
	PROPER LEVEL	

Note :

INSPECTION BY

APPROVED BY

QA/GL

QA COORDINATOR